DO NOW — RECALL

# Character sets: ASCII & Unicode

Name: _____ Class: _____ Date: _____

**1. Which data type should store a telephone number beginning with 0? Explain briefly.**

**2. Convert $41_{16}$ to denary and eight-bit binary.**

**3. Apply AND 00001111 to 10100101. Explain which bits the mask preserves.**

**4. In 6+4 two's-complement floating-point format, subtract 010000 0010 from 011000 0010. Give a normalised result.**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

String: preserve the leading zero; it is an identifier, not a quantity for arithmetic.

### 2. Model answer

65; 01000001.

### 3. Model answer

00000101. The low four bits are preserved; the high four are cleared.

### 4. Model answer

3 − 2 = 1 → 010000 0001.

### By the end, you can…

• explain how character codes represent text;

• compare ASCII and Unicode and choose an appropriate encoding;

• apply character-code calculations and calculate storage using stated encoding rules.

### Vocabulary — know these terms

Recall the term — Character set

Recall the term — Code point

Recall the term — Encoding

Recall the term — ASCII

Recall the term — Unicode

Recall the term — UTF-8

Recall the term — Control character

Recall the term — Repertoire

NEW LEARNING

# Read, annotate and explain

## Useful ASCII anchors

'A' = 65 'Z' = 90
'a' = 97 'z' = 122
'0' = 48 '9' = 57
space = 32

Standard ASCII uses 7 bits → 128 possible codes.

A character code represents text. The character '7' is not the same value as the integer 7 .

## Characters need agreed codes

A character set defines characters and their assigned numeric codes. An encoding specifies how those codes are represented as bytes.

## Standard ASCII is limited

Seven bits provide $2^7 = 128$ codes for letters, digits, punctuation and control characters.

Eight-bit extensions can provide 256 values, but different extensions may map the extra codes differently.

## Unicode supports a much larger repertoire

Unicode assigns code points to characters from many writing systems and symbols. Its first 128 code points match ASCII.

## Use the encoding stated in storage questions

Storage depends on the number of code points and the number of bytes used for each one. Do not assume every Unicode character uses the same width.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

## Encode the text “AB”

### Start with the character codes

```
A = 65
B = 66
```

ASCII assigns a numeric code to each character.

### Write the 7-bit ASCII values

```
A = 1000001
B = 1000010
```

### Add a leading zero

```
A → 01000001
B → 01000010

AB → 01000001 01000010
```

The leading zero lets each 7-bit ASCII code sit inside an 8-bit byte; it does not expand the standard ASCII repertoire.

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

### Guided 1 — TRY: 2. Character versus quantity

Use the I Do method and build the answer before checking.

Method / working:

### Guided 2 — TRY: 3. Use a code range

Use the I Do method and build the answer before checking.

Method / working:

### Guided 3 — TRY: 4. Choose an encoding

Use the I Do method and build the answer before checking.

Method / working:

### Guided 4 — TRY: 5. Calculate storage from stated widths

Use the I Do method and build the answer before checking.

Method / working:

### Guided 5 — TRY: OCR JUNE 2022 · Q5(e)(ii–iii) · [3]

Use the I Do method and build the answer before checking.

Method / working:

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

## Question 1

Question 1 · ADAPTED OCR PROGRAMMING TASK · [3]
Complete the three gaps to count uppercase ASCII letters. asc returns a character's code. A = 65 and Z = 90. The loop examines each character once.
CODE:

```
function countCapitals(text)
count = 0
for each character in text
code = asc(character)
if code >= 65 _____ code <= 90 then
count = _____
endif
next character
return _____
endfunction
```

Adapted from OCR June 2022 Q5(e)(i): simplified loop and changed gaps; classroom allocation of one mark per gap. OCR June 2022, Q5(e), page 24

## Question 2

Question 2 · ADAPTED EXAM COMPARISON · [4]
A school app must store English and Chinese messages. Explain why standard ASCII is unsuitable [1], why Unicode is suitable [1], give one similarity between them [1], and explain why the receiver must know the encoding [1].
Contextual adaptation of OCR June 2022 Q5(e)(ii–iii); classroom marks.

## Question 3

Question 3 · ORIGINAL EXAM-STYLE CALCULATIONS · [6]
Use A = 65, a = 97 and 0 = 48. Show a code calculation and an eight-bit byte for each: (a) C, (b) c, (c) the digit character 5. [2 each]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

## Question 4

Question 4 · ORIGINAL CAPACITY QUESTION · [3]
State how many codes standard seven-bit ASCII provides [1]. Calculate the minimum number of bits needed for a fixed-width character set containing 300 distinct codes, showing why the next smaller width is insufficient. [2]

## Question 5

Question 5 · ORIGINAL ENCODING APPLICATION · [4]
A UTF-8 string contains 20 ASCII code points, three euro-sign code points and two emoji code points. The byte widths are 1, 3 and 4 respectively. Ignore metadata. Calculate bytes [2] and bits [1]. Explain why multiplying the number of code points by 2 bytes is not valid here. [1]

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking.

**Standard ASCII provides 128 character codes.**

**Circle: Confident / Getting there / Need help**

Explain what you can now do independently.

## GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

# Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

### Guided 1 — CHECK

Build the answer

### Guided 2 — CHECK

Build the answer
For lowercase c : 97 + 2 = 99; uppercase C = 99 − 32 = 67.

### Guided 3 — CHECK

Build the answer
Both ends must interpret the bytes using the same encoding.

### Guided 4 — CHECK

Build the answer

### Guided 5 — CHECK

Build the answer